

# Customer Shopping Behavior Analysis

Uncovering spending patterns, customer segments, and product preferences from 3,900 purchases to guide strategic business decisions.



# The Dataset at a Glance

Transactional data spanning customer demographics, purchase details, and shopping behavior across four product categories.

 37 missing values in the Review Rating column, imputed by category median.

**3.9K**

Purchases

**18**

Columns

**4**

Categories

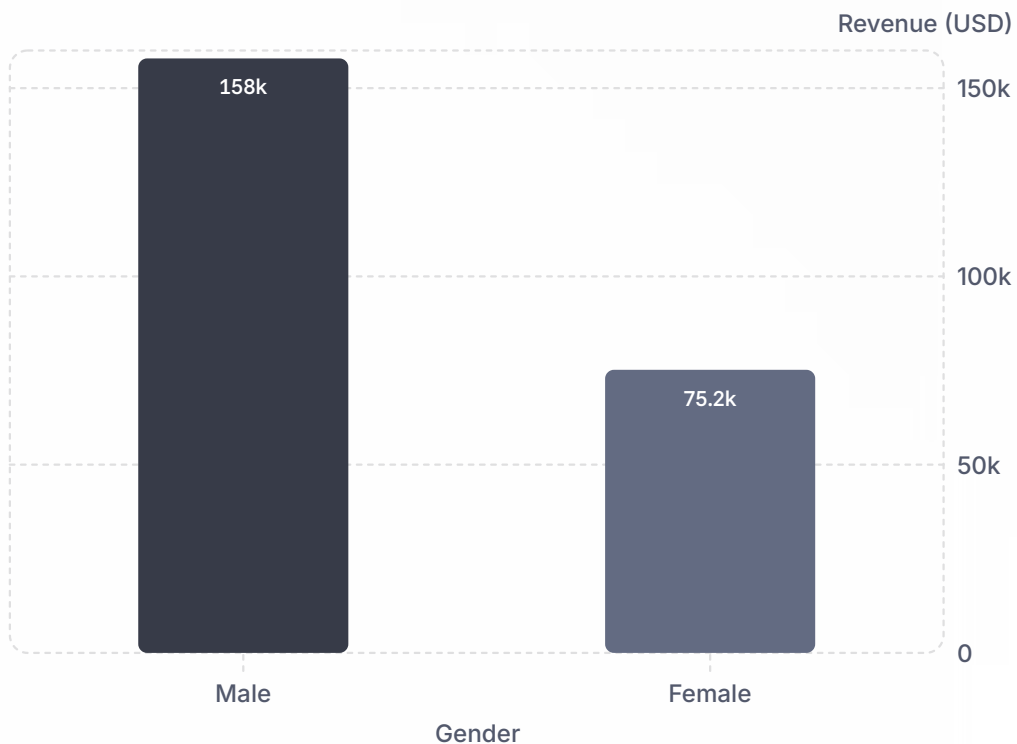
## Step 1: Data Preparation

# Cleaning with Python



Dropped redundant `promo_code_used`, then connected Python to PostgreSQL for SQL analysis.

# Revenue by Gender



## Male customers drive revenue

Male shoppers generated more than double the revenue of female shoppers — a key signal for targeted campaigns.

# Subscribers vs. Non-Subscribers

## Subscribers

1,053 customers

\$59.49 avg spend

\$62,645 revenue

## Non-Subscribers

2,847 customers

\$59.87 avg spend

\$170,436 revenue

Only 27% of customers subscribe - a major growth opportunity.

# Products & Ratings Insights

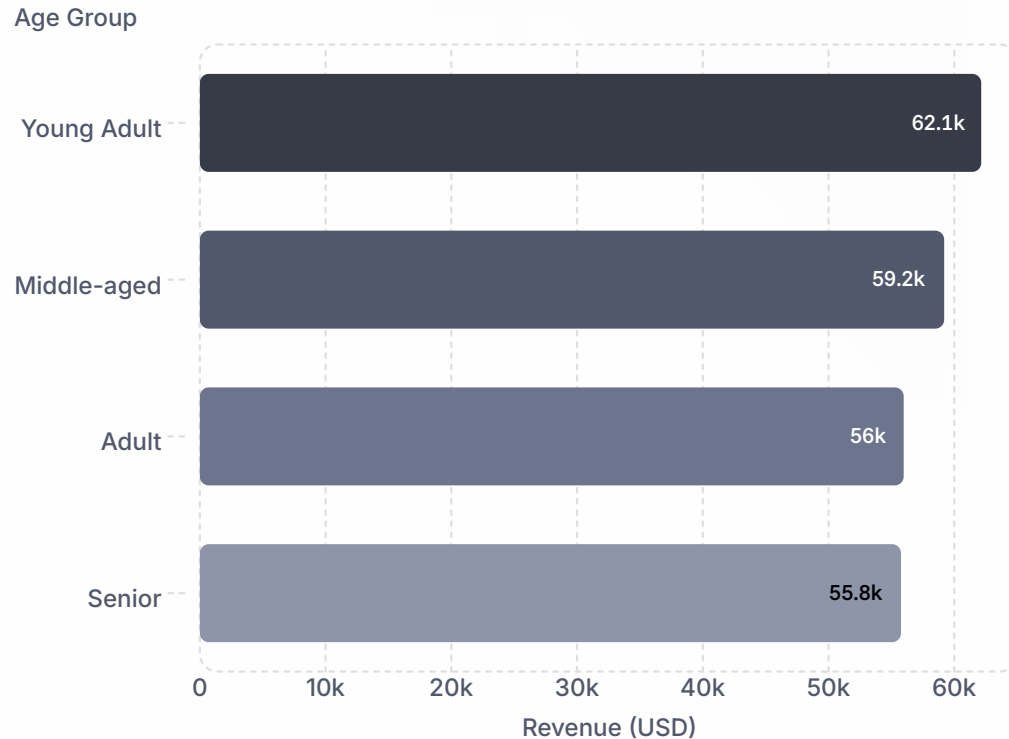
## Top Rated Products

|         |      |
|---------|------|
| Gloves  | 3.86 |
| Sandals | 3.84 |
| Boots   | 3.82 |
| Hat     | 3.80 |
| Skirt   | 3.78 |

## Most Discount-Dependent

|          |       |
|----------|-------|
| Hat      | 50.0% |
| Sneakers | 49.7% |
| Coat     | 49.1% |
| Sweater  | 48.2% |
| Pants    | 47.4% |

# Revenue by Age Group

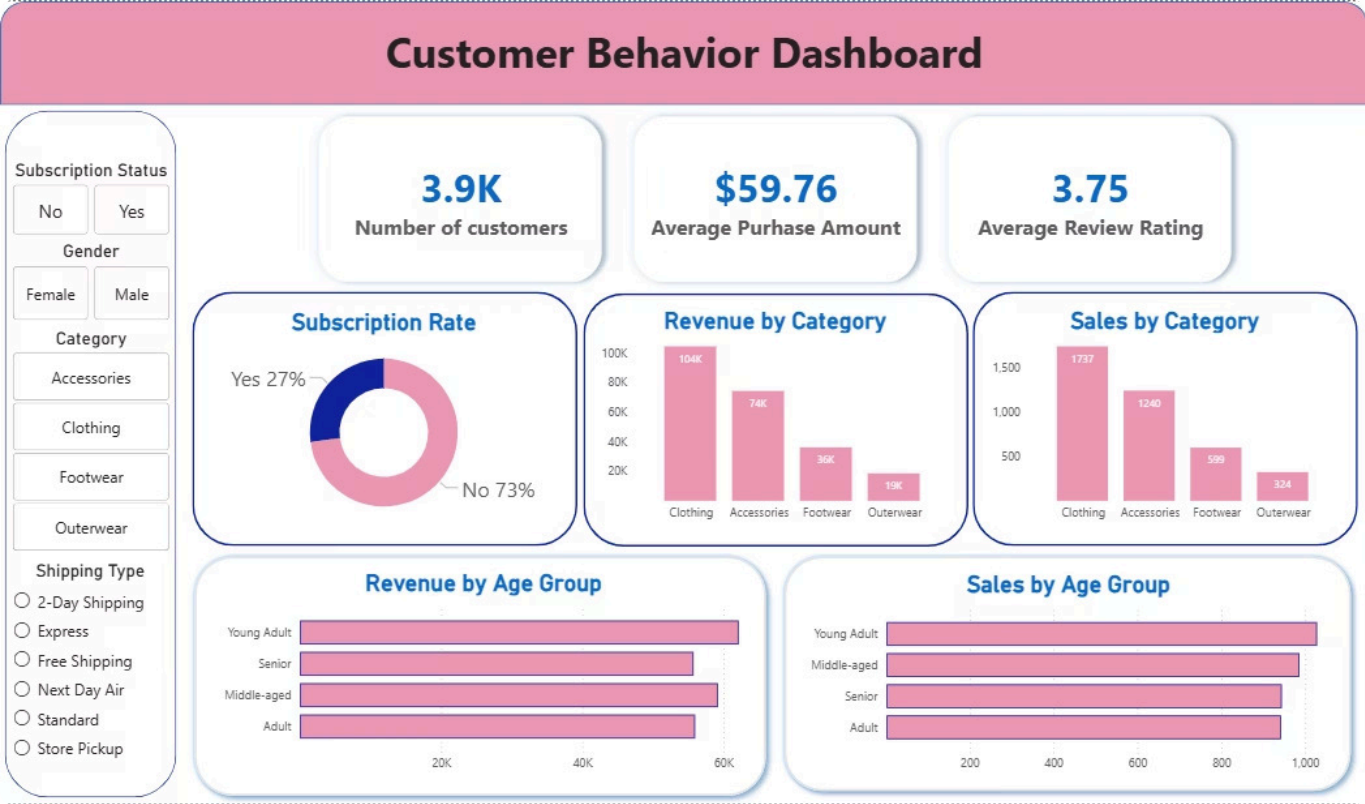


## Young Adults buy the most

Revenue is evenly spread, but Young Adults contribute the most- a prime target for focused marketing.

📄 Express shipping averaged \$60.48 vs. \$58.46 for Standard.

# Power BI Dashboard



## Interactive Insights

An interactive dashboard surfaces key metrics with filters for subscription status, gender, category, and shipping type.

- 3.9K customers
- \$59.76 average purchase
- 3.75 average rating



# Business Recommendations

## → Boost Subscriptions

Promote exclusive benefits for subscribers.

## → Loyalty Programs

Reward repeat buyers into the "Loyal" segment.

## → Review Discounts

Balance sales boosts with margin control.

## → Targeted Marketing

Focus on high-revenue age groups and express users.

